

Executive Summary

Low-latency anonymity systems like Tor [1] encrypt content, but they still leak traffic patterns that let Website Fingerprinting (WF) attacks [2] identify which page a user visits. Tamaraw [3] is designed to defend against this leakage by adding padding and sending packets at fixed intervals, but doing so increases bandwidth use, creating a trade-off between Success-Attack Rate (SAR) and Bandwidth Overhead (BO). In previous studies, the investigation of the trade-offs is limited to some certain fixed BO, and the understanding for a wide range of BO is still lacking.

In this project, we evaluate the trade-off for the one-page setting [4] by sweeping Tamaraw's Pad Length (PadL) parameter [3], which controls BO, and measuring how four modern attack schemes, namely, Deep Fingerprinting (DF), Tik-Tok, Robust Fingerprinting (RF), and Var-CNN [2] perform against the defended traffic. Our results show that while a higher BO reduces SAR, all four attack schemes [2] remain well above random guessing, confirming that stronger privacy is possible, but comes at a BO cost.

Backgrounds and Goal

WF attacks infer visited websites from encrypted traffic patterns, e.g., timing, direction, volume [2]. Tamaraw masks these patterns using fixed-interval transmissions and padding at the cost of extra BO [3]. Prior studies examine trade-offs only at fixed BO, leaving broader BO behavior unclear.

In this study, we focus on the one-page setting [4], which is a stricter evaluation compared to the multiple-page setting because the attacker only needs to identify one sensitive page instead of choosing among many. That better matches targeted privacy threats, where even one detected visit can matter. Our main goal is to evaluate how Tamaraw [3] performs in the one-page setting [4] by measuring how much BO is needed to reduce SAR, consisting of accuracy — overall correct guesses, and F1 score — how well it balances catching true pages and avoiding false alarms, against modern attacks on contemporary datasets.

Experimental Plan

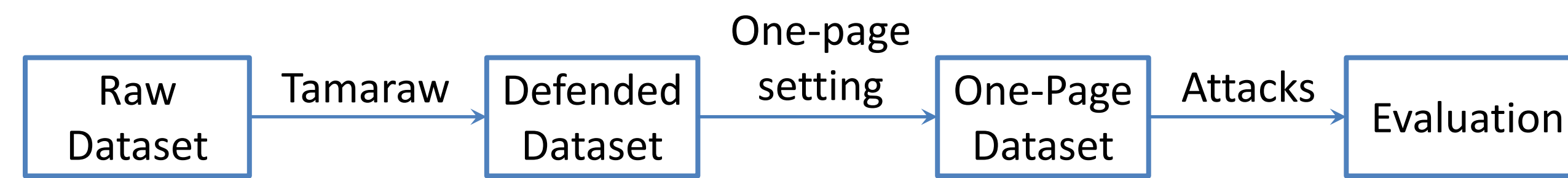


Fig. 1: Experiment pipeline

Table 1: WF Attacks and how they work

Deep Fingerprinting (DF)	Tik-Tok	Robust Fingerprinting (RF)	Var-CNN
Direction	Direction + Timing	Direction + Timing with CNN features	TAM features

In our Closed-World (CW) dataset [2], each traffic trace from 95 monitored pages is passed through Tamaraw [3], which pads packets for each packet group via the PadL parameter. Our PadL sweep corresponds to roughly 100% to 400% BO, which is computed as the ratio of defended packets to original packets (defended data size ÷ raw data size minus one, as a percentage).

For each PadL, a one-page dataset is built with 95 monitored pages and split into train, validation, and test sets. Four attack schemes are then trained on the defended data: DF, Tik-Tok, Var-CNN, and RF [2]. For each model, accuracy and F1 score are measured and then matched with the corresponding BO.

Main Results

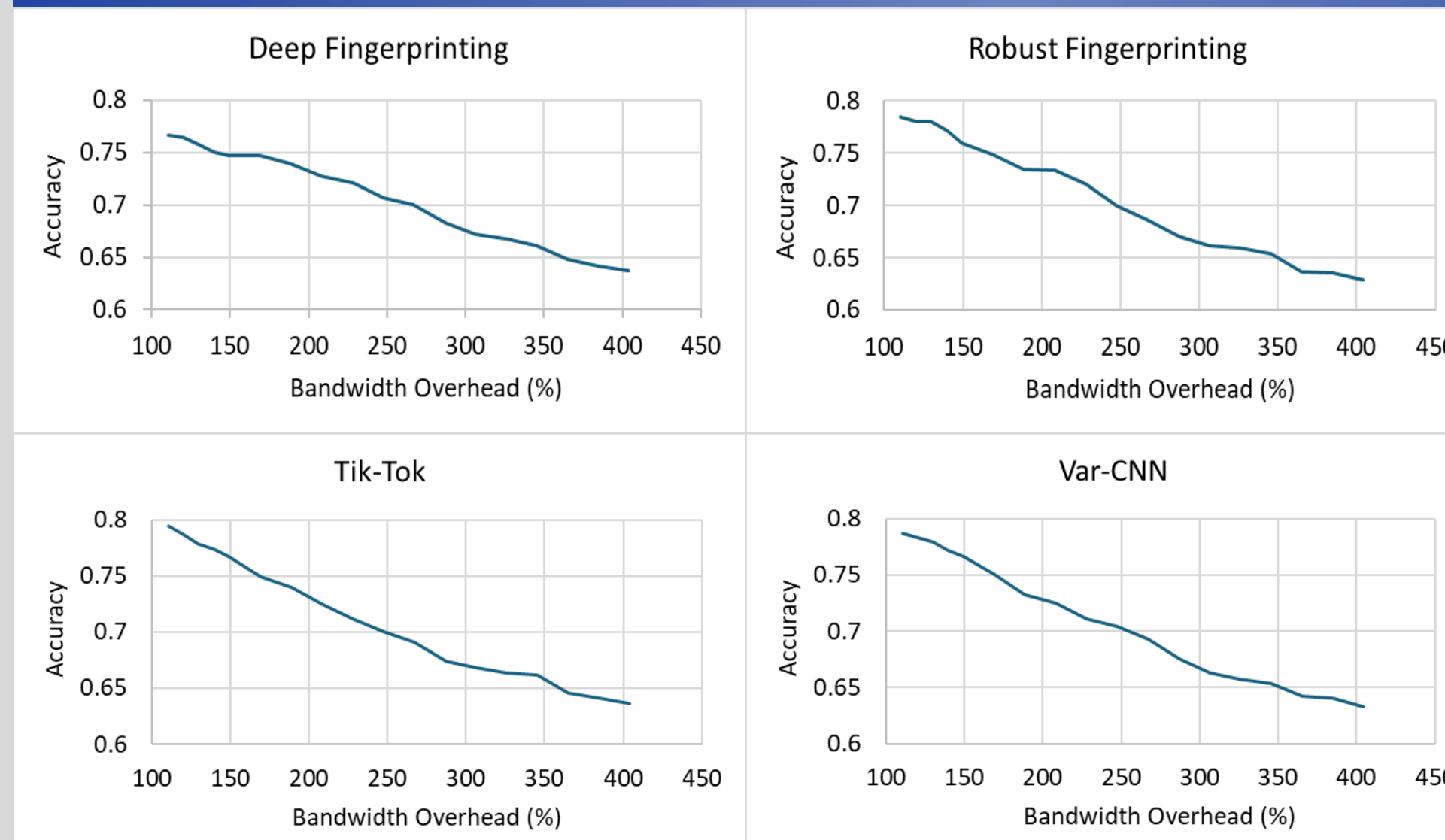


Fig. 2: Accuracy vs. bandwidth overhead for tamaraw under four WF attack schemes.

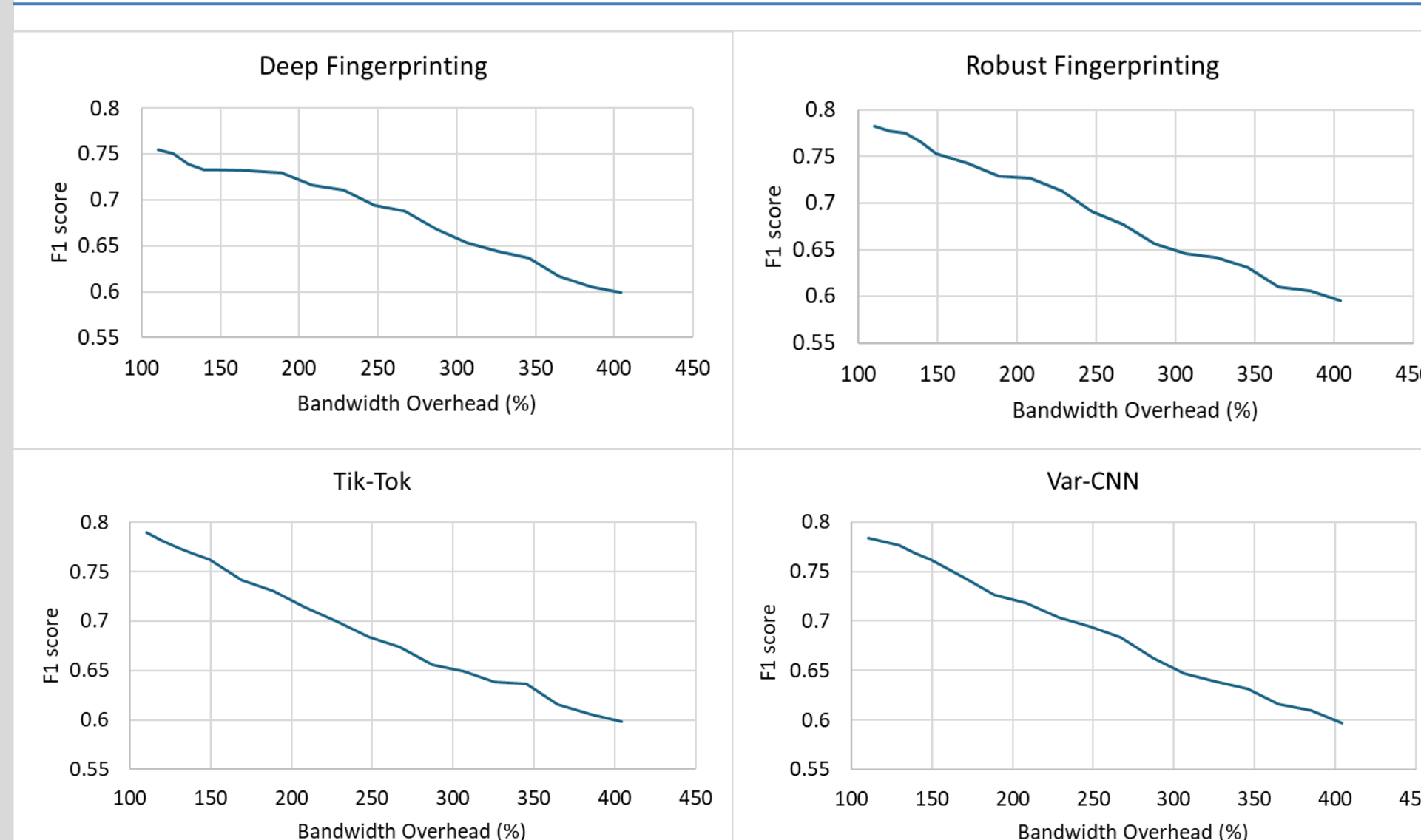


Fig. 3: F1 score vs. bandwidth overhead for Tamaraw under four WF attack schemes.

Discussions

- As PadL increases, attack accuracy and F1 score both decline, but the decrease is smooth and modest rather than sudden, so the models still perform well above random even at our highest overheads.
- This behavior reveals a strong privacy–cost trade-off: to noticeably hurt attack performance, Tamaraw must add a large amount of padding, which substantially increases bandwidth use and would be costly to deploy in practice.
- The four WF attack schemes [2] react differently to Tamaraw [3] — some lose accuracy faster than others—but all remain surprisingly strong over much of the PadL range, showing that Tamaraw [3] weakens modern attacks in the one-page setting [4] but does not fully prevent page identification.

Conclusions and Future Work

We studied the trade-off between accuracy/F1 score and bandwidth overhead for WF attacks [2].

- Accuracy: As protection (PadL and overhead) increases, attack accuracy steadily goes down, but it remains well above random even at our highest overheads.
- F1 score: F1 score also drops with more protection, meaning the attacks both miss more true pages and raise more false alarms, yet they still identify the monitored page with fairly high confidence.

One possible direction is to change other Tamaraw [3] parameters (such as timing and padding rules) or combine Tamaraw with additional randomness to see if accuracy and F1 score can be pushed much lower without adding too much extra BO.

References

- [1] R. Dingledine, N. Mathewson, and P. Syverson, “Tor: the second-generation onion router,” in *Proc. USENIX Secur. Symp., ser. SSYM’04*, 2004, p. 21.
- [2] X. Deng, Q. Li, and K. Xu, “Robust and reliable early-stage website fingerprinting attacks via spatial-temporal distribution analysis,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2024.
- [3] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, “A systematic approach to developing and evaluating website fingerprinting defenses,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2014, p. 227–238.
- [4] T. Wang, “The one-page setting: A higher standard for evaluating website fingerprinting defenses,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2021, p. 2794–2806.

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